

MCU



Load Capacitance : 6pF ~ 12pF

b. Capacitor & Resistor Requirements (AI Generated)

- C2 & C3
Capacitance : 6pF ~ 12pF
Voltage Rating : 10V or higher
Tolerance : +- 5% or lower
Minimum package : 0402
Type : C0F or NP0
Placement : near X1
- C1, C4, C5, C32
Capacitance : 100nF
Voltage Rating : 16V or higher
Tolerance : +- 10%
Minimum package : 0402
Type : X7R or X5R
Placement : Near designated pin
- C33
Capacitance : 4.7uF
Voltage Rating : 10V or higher
Tolerance : +- 10%
Minimum package : 0603
Type : X7R or X5R
Placement : Near power pin
- R13
Resistance : 100Ω
Tolerance : +- 5%
Power Rating : 1/16W
Minimum package : 0402
- R17
Resistance : 10kΩ
Tolerance : +- 5%
Power Rating : 1/16W
Minimum package : 0402

c. STM32 Pin Data

Source: Table 15. From STM32L031x4/6 Datasheet

Pin No.	Pin Name	Pin Type	Alternate/Additilonal Functions	Assigned As
1	VDD	S	-	VCC
2	PC14 - OSC32 IN	I/O	-	OSC_IN
3	PC15 - OSC32 OUT	I/O	-	OSC_OUT
4	NRST	I/O	-	NRST
5	VDDA	S	-	AVCC

6	PA0 - CK_IN	I/O	COMP1_INM6, ADC_IN0, RTC_TAMP2/WKUP1, LPTIM1_IN1, TIM2_CH1, USART2_CTS, TIM2_ETR, COMP1_OUT	UV_SIG (ANALOG INPUT: ADC_IN0)
7	PA1	I/O	EVENTOUT, LPTIM1_IN2, TIM2_CH2, I2C1_SMBA, USART2_RTS, TIM21_ETR, COMP1_INP, ADC_IN1	UV_EN (DIGITAL OUTPUT)
8	PA2	I/O	TIM21_CH1, TIM2_CH3, USART2_TX, LPUART1_TX, COMP2_OUT, COMP2_INM6, ADC_IN2, RTC_TAMP3/RTC_TS/ RTC_OUT/WKUP3	STM32_TX (LPUART1_TX)
9	PA3	I/O	TIM21_CH2, TIM2_CH4, USART2_RX, LPUART1_RX, COMP2_INP, ADC_IN3	STM32_RX (LPUART1_RX)
10	PA4	I/O	SPI1_NSS, LPTIM1_IN1, USART2_CK, TIM22_ETR COMP1_INM4, COMP2_INM4, ADC_IN4	EPD_BUSY (DIGITAL INPUT)
11	PA5	I/O	SPI1_SCK, LPTIM1_IN2, TIM2_ETR, TIM2_CH1, COMP1_INM5, COMP2_INM5, ADC_IN5	EPD_CLK (SPI1_SCK)
12	PA6	I/O	SPI1_MISO, LPTIM1_ETR, LPUART1_CTS, TIM22_CH1, EVENTOUT, COMP1_OUT, ADC_IN6	EPD_CS (OUTPUT)
13	PA7	I/O	SPI1_MOSI, LPTIM1_OUT, USART2_CTS, TIM22_CH2, EVENTOUT, COMP2_OUT, ADC_IN7	EPD_DIN (SPI1_MOSI)
14	PB0	I/O	EVENTOUT, SPI1_MISO, USART2_RTS, TIM2_CH3 ADC_IN8, VREF_OUT	EPD_DC (OUTPUT)
15	PB1	I/O	USART2_CK, SPI1_MOSI, LPUART1_RTS, TIM2_CH4, ADC_IN9, VREF_OUT	EPD_RST (OUTPUT)
16	VSS	S	-	GND
17	VDD	S	-	VCC
18	PA8	I/O	MCO, LPTIM1_IN1, EVENTOUT, USART2_CK, TIM2_CH1	EPD_EN (OUTPUT)
19	PA9	I/O	MCO, I2C1_SCL, USART2_TX,	I2C_SCL

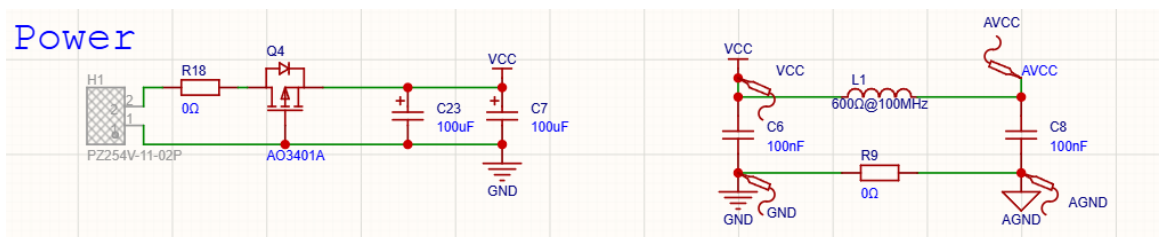
			TIM22_CH1	
20	PA10	I/O	I2C1_SDA, USART2_RX, TIM22_CH2	I2C_SDA
21	PA13	I/O	SWDIO, LPTIM1_ETR, LPUART1_RX	SWDIO
22	PA14	I/O	SWCLK, LPTIM1_OUT, I2C1_SMBA, USART2_TX, LPUART1_TX	SWCLK
23	PA15	I/O	SPI1_NSS, TIM2_ETR, EVENTOUT, USART2_RX, TIM2_CH1	No Connection
24	PB3	I/O	SPI1_SCK, TIM2_CH2, EVENTOUT, COMP2_INN	LED (DIGITAL OUTPUT)
25	PB6	I/O	USART2_TX, I2C1_SCL, LPTIM1_ETR, TIM21_CH1, COMP2_INP	BTN_2 (DIGITAL INPUT)
26	PB7	I/O	USART2_RX, I2C1_SDA, LPTIM1_IN2, COMP2_INP, VREF_PVD_IN	BTN_1 (DIGITAL INPUT)
27	BOOT0	I	-	BOOT0
28	VSS	S	-	GND

Note:

Pin Type:

- S: Supply Pin
- I: Input Only
- O: Output Only
- I/O: Input Output
- B: Boot0 Pin
- RST: Bidirectional reset pin

2. Power Circuitry



a. Electrical Rating Requirements

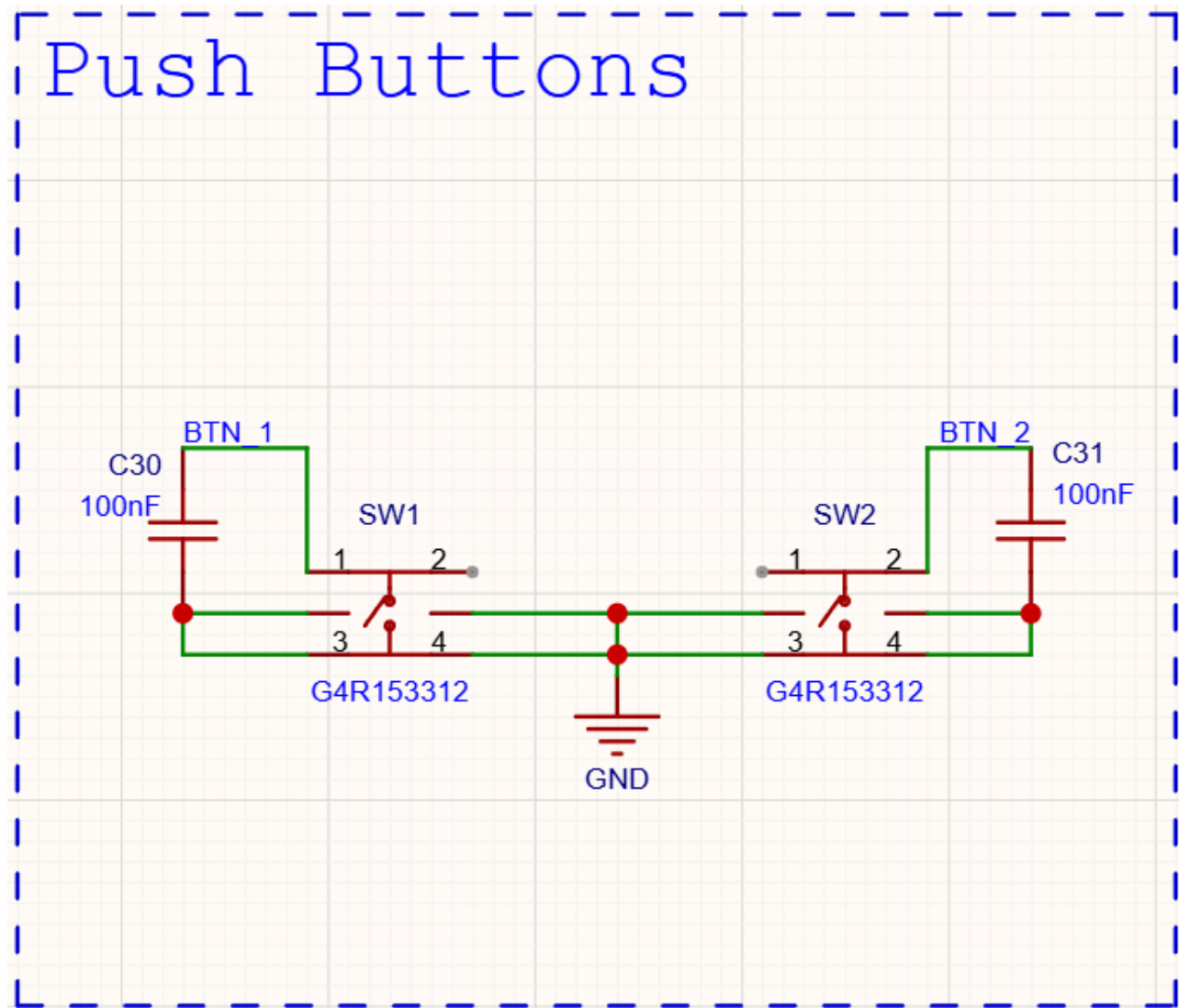
- Power Source : CR2032 Battery

- Voltage : 3V
- Internal R : $15\Omega \sim 40\Omega$
- Capacity : 200 mAh ~ 250 mAh
- Absolute Maximum
 - Voltage : 3.5V
 - Current : 3mA

b. Components Specification Requirements (AI Generated)

- L1 (Ferrite Bead)
 - Impedance : $600\Omega @ 100 \text{ MHz}$
 - Current Rating : 200mA ~ 500mA
 - Minimum package : 0603
- C6, C8
 - Capacitance : 100nF
 - Voltage Rating : 16V or higher
 - Tolerance : $\pm 10\%$
 - Minimum package : 0402
 - Type : X7R or X5R
 - Placement : Near ferrite bead
- C23, C7
 - Capacitance : 100uF
 - Voltage Rating : 10V or higher
 - Tolerance : $\pm 20\%$
 - Minimum package : A-Case or similar
 - Type : Tantalum
 - Placement : Near ferrite bead
- R6, R18
 - Package : 0605

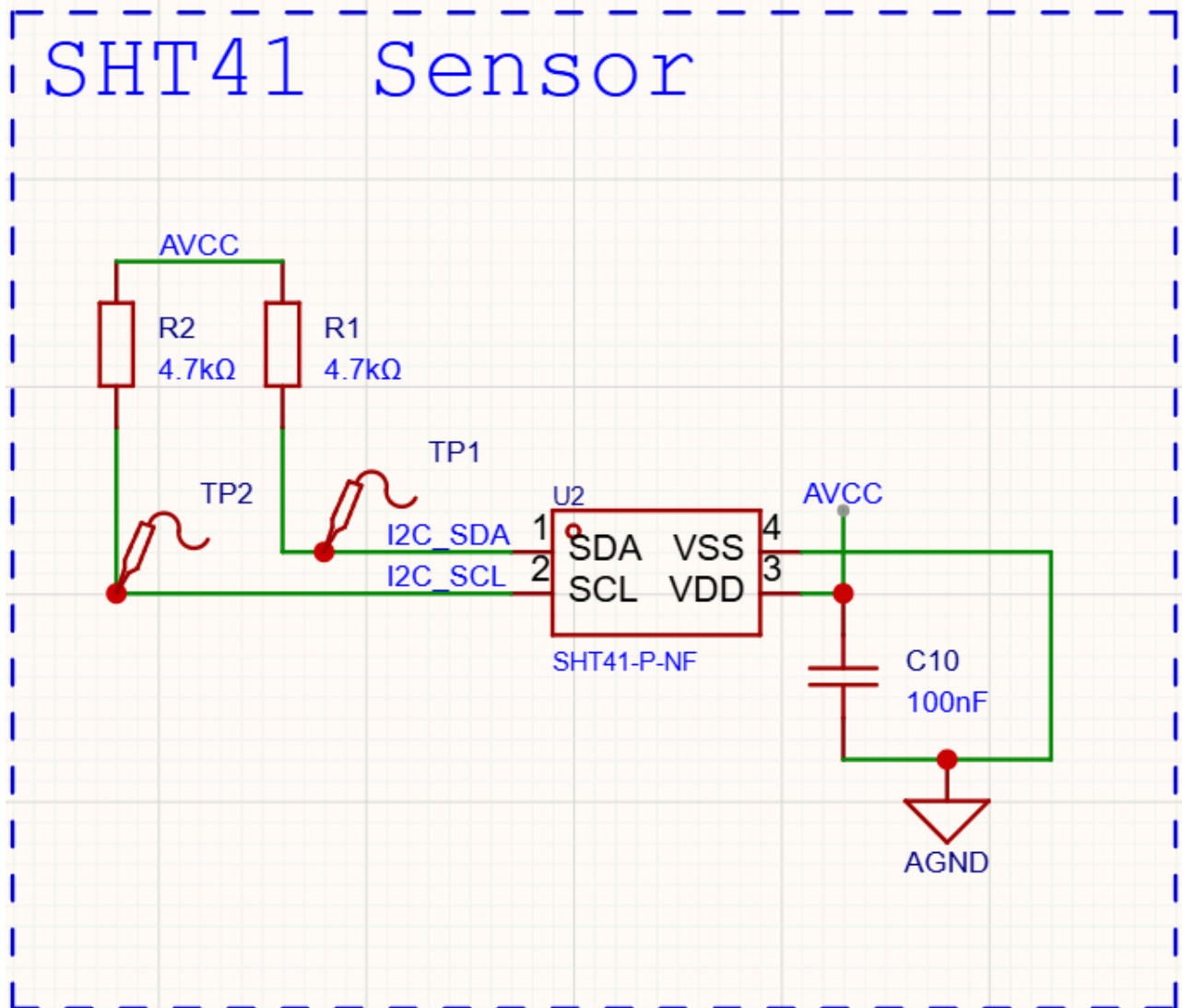
3. Push Button



a. Components Specification Requirements (AI Generated)

- SW1, SW2
 - Minimum cap Ø : 2mm
 - Package : SMD
- C30, C31
 - Capacitance : 100nF
 - Voltage Rating : 16V or higher
 - Tolerance : +- 10%
 - Minimum package : 0402
 - Type : X7R or X5R
 - Placement : Near designated pin

4. SHT41



a. Datasheet Informations

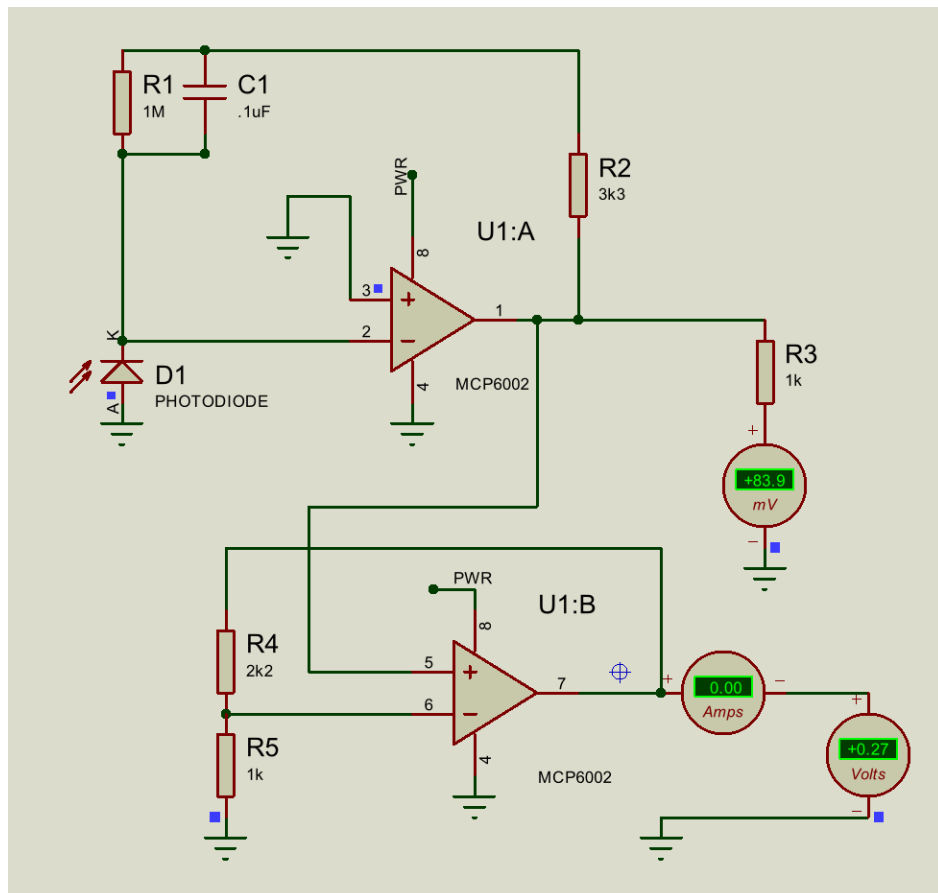
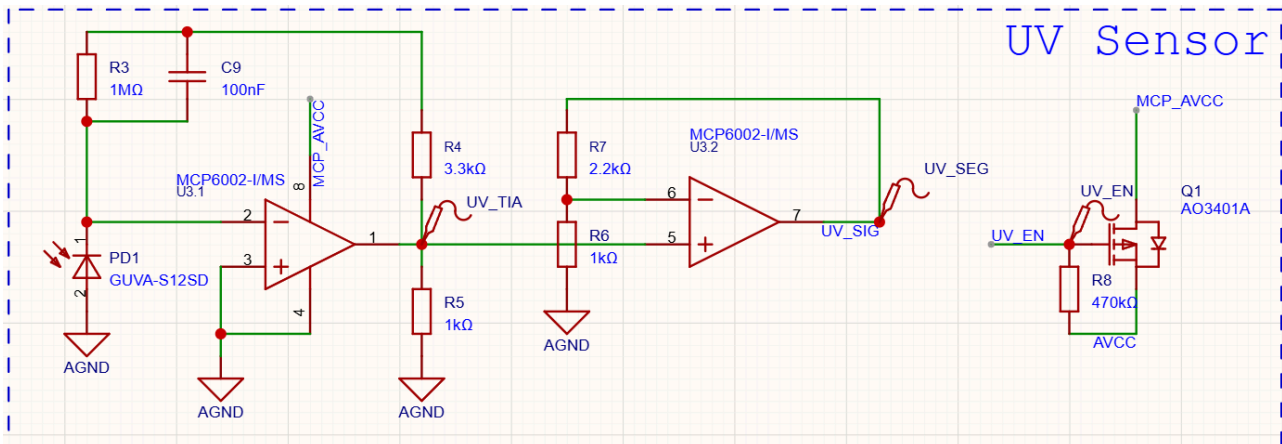
- SHT41
 - Supply Voltage : 1.08V ~ 3.6V
 - Temperature : -40°C ~ 125°C
 - Current : 80 nA ~ 0.4 uA
 - Interface : I2C
 - I2C Address : 0x44/0x45/0x46

b. Components Specification Requirements (AI Generated)

- R1, R2
 - Resistance : 4.7kΩ
 - Tolerance : +- 5%
 - Power Rating : 1/16W
 - Minimum package : 0402
- C10
 - Capacitance : 100nF
 - Voltage Rating : 16V or higher

Tolerance : +/- 10%
 Minimum package : 0402
 Type : X7R or X5R
 Placement : Near the power pin

5. UV Sensor



GUVA-S12SD Simulation Data

a. Datasheet Informations

- GUVA-S12SD
 Spectral Range : 240 nm ~ 370 nm
 Photo Current : 21 nA / 1 UVI
 Soldering Temp : 260°C

- MCP6002
 - Supply Voltage : 1.8V ~ 6.0V
 - Operating Temp : -40°C ~ 85°C
 - Package : MSOP-8
 - Quiescent Current : 100 uA ~ 320 uA
 - Soldering Temp : 260°C
- AO3401A
 - MOSFET Type : P-Channel
 - Quiescent Current : 0.1uA ~ 1uA

b. Components Specification Requirements (AI Generated)

- C9
 - Capacitance : 100nF
 - Voltage Rating : 16V or higher
 - Tolerance : +- 10%
 - Minimum package : 0402
 - Type : X7R or X5R
 - Placement : Near photodiode
- R3
 - Resistance : 1M Ω
 - Tolerance : +- 1%
 - Power Rating : 1/16W
 - Minimum package : 0402
- R8
 - Resistance : 470k Ω
 - Tolerance : +- 5%
 - Power Rating : 1/16W
 - Minimum package : 0402
- Post-TIA Level Shifting
 - R4
 - Resistance : 3.3k Ω
 - Tolerance : +- 1%
 - Power Rating : 1/16W
 - Minimum package : 0402
 - R5
 - Resistance : 1k Ω
 - Tolerance : +- 1%
 - Power Rating : 1/16W
 - Minimum package : 0402
- Op-Amp Gain
 - R6
 - Resistance : 1k Ω
 - Tolerance : +- 1%
 - Power Rating : 1/16W
 - Minimum package : 0402
 - R7
 - Resistance : 2.2k Ω
 - Tolerance : +- 1%
 - Power Rating : 1/16W
 - Minimum package : 0402

c. Op-Amp Configuration Requirements:

Maximum output voltage : 1.2V (STM32 VREFINT)
TIA : 21 nA -> 21mV
Op-Amp Gain : 3.2x: 21mV -> 0.07V

[GUVA-S12SD Simulation Data - Google Sheets](#)

6. USB Interface

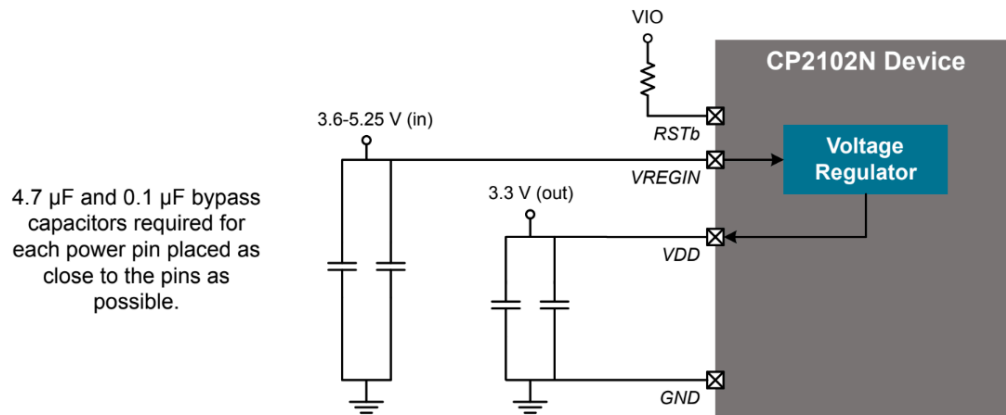
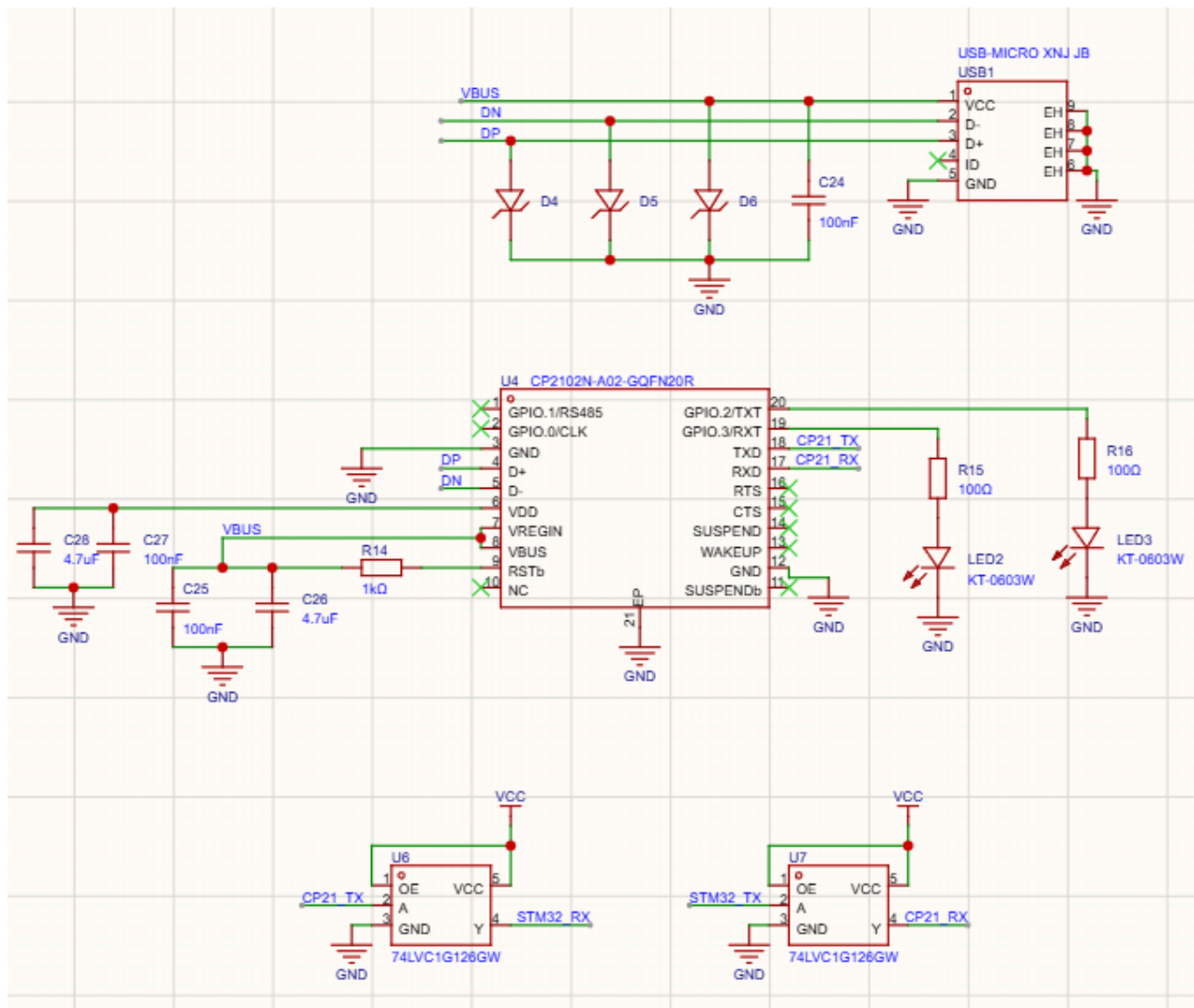


Figure 2.2. Connection Diagram with Voltage Regulator Used and USB Connected (Self-Powered)

Typical Application (Source: CP2102N Datasheet)



a. Datasheet Informations (CP2102N)

Part Number : CP2102N-A02-GQFN20R
 Supply Volt (VREGIN): 3.6V to 5.25V (from USB VBUS)
 Regulated VOUT : 3.3V (Internal Regulator Output)
 Operating Temp : -40°C ~ 85°C
 Comm. Interface : USB 2.0 Full-Speed (12 Mbps) to UART Bridge

b. Components Specification Requirements (AI Generated)

- USB1
 - Conn. Type : Micro-B USB Receptacle
 - Placement : Boardge edge
- C24, C25, C27
 - Capacitance : 100nF
 - Voltage Rating : 16V or higher
 - Tolerance : +- 10%
 - Minimum package : 0402
 - Type : X7R or X5R
 - Placement : Adjacent to VBUS and VDD
- C26, C28
 - Capacitance : 4.7uF
 - Voltage Rating : 16V or higher
 - Tolerance : +- 10%
 - Minimum package : 0805
 - Type : X7R or X5R
 - Placement : Adjacent to VBUS and VDD

- C21, C22
Capacitance : 4.7uF
Voltage Rating : 50V or higher
Tolerance : +- 10%
Minimum package : 0805
Type : X7R or X5R

- C11~C20
 - Capacitance : 1uF
 - Voltage Rating : 16V or higher
 - Tolerance : +- 10%
 - Minimum package : 0402
 - Type : X7R or X5R
 - Placement : Adjacent to corresponding FPC connector
- R10
 - Resistance : 3 Ω
 - Tolerance : +- 5%
 - Power Rating : 1/4W
 - Minimum package : 0603
- R11
 - Resistance : 470k Ω
 - Tolerance : +- 5%
 - Power Rating : 1/16W
 - Minimum package : 0402
- R12
 - Resistance : 10k Ω
 - Tolerance : +- 5%
 - Power Rating : 1/16W
 - Minimum package : 0402
- D1, D2, D3
 - Part Number : MBR0530
 - Reverse Voltage : 30V or higher
 - Forward Current : 0.5A
 - Package : SOD-123
- Q2 (AO3401A)
 - MOSFET Type : P-Channel
 - Quiescent Current : 0.1uA ~ 1uA
- Q3
 - Part Number : SI1304BDL
 - Voltage : 20v to 30v

b. Interface Mapping

i. FPC-24

Pin No.	Datasheet Name ¹	Reference Name ²	Reference Connection ²	Current Connection ³
1	NC	HLT_CTL	NC	NC
2	GDR	GDR	Q3 Gate	Q3 Gate
3	RESE	RESE	Q3 Source	Q3 Source
4	NC	VGL	CAP	CAP
5	VSH2	VGH	CAP	CAP
6	TSCL	TSCL	NC	NC

7	TSDA	TSDA	NC	NC
8	BS1	BS	Solder Bridge	Solder Bridge
9	BUSY	BUSY	GPIO	PA4 (10)
10	RES#	RST	GPIO	PB1 (15)
11	D/C#	D/C	GPIO	PB0 (14)
12	CS#	CS	GPIO	PA6 (12)
13	SCL	SCLK	GPIO (SPI)	PA5 (11)
14	SDA	SDI	GPIO (SPI)	PA7 (13)
15	VDDIO	VDDIO	3V Power	EPD_VDD
16	VC1	VC1	3V Power	EPD_VDD
17	VSS	VSS	GND	GND
18	VDD	VDD	CAP	CAP
19	VPP	VPP	CAP	CAP
20	VSH	VSH	CAP	CAP
21	VGH	PREVGH	Diode	D3
22	VSL	VSL	CAP	CAP
23	VGL	PREVGL	Diode	D2
24	VCOM	VCOM	CAP	CAP

¹ from [waveshare 2.13 e-Paper V4 Specification](#)

² from reference design

³ on this implementation (v1.2)